Class Prototype Alignment Concentrates Chest X-ray Representations in Federated and Pooled Training

Federated and pooled training

Jingjing Yang and Xia Zhao
Hebei North University

Geometry changes reliably; prediction must be checked separately.

One class mean is compact, but it is also a strong constraint

Mechanism under test

The alignment term pulls images with the same label toward one class mean, even when that label contains diverse anatomy and acquisition variation.

Matched controls

ChestX-ray14
Pneumonia vs No Finding
2 client allocations, 3 K values, 3 seeds
pooled and VinDr-CXR geometry checks

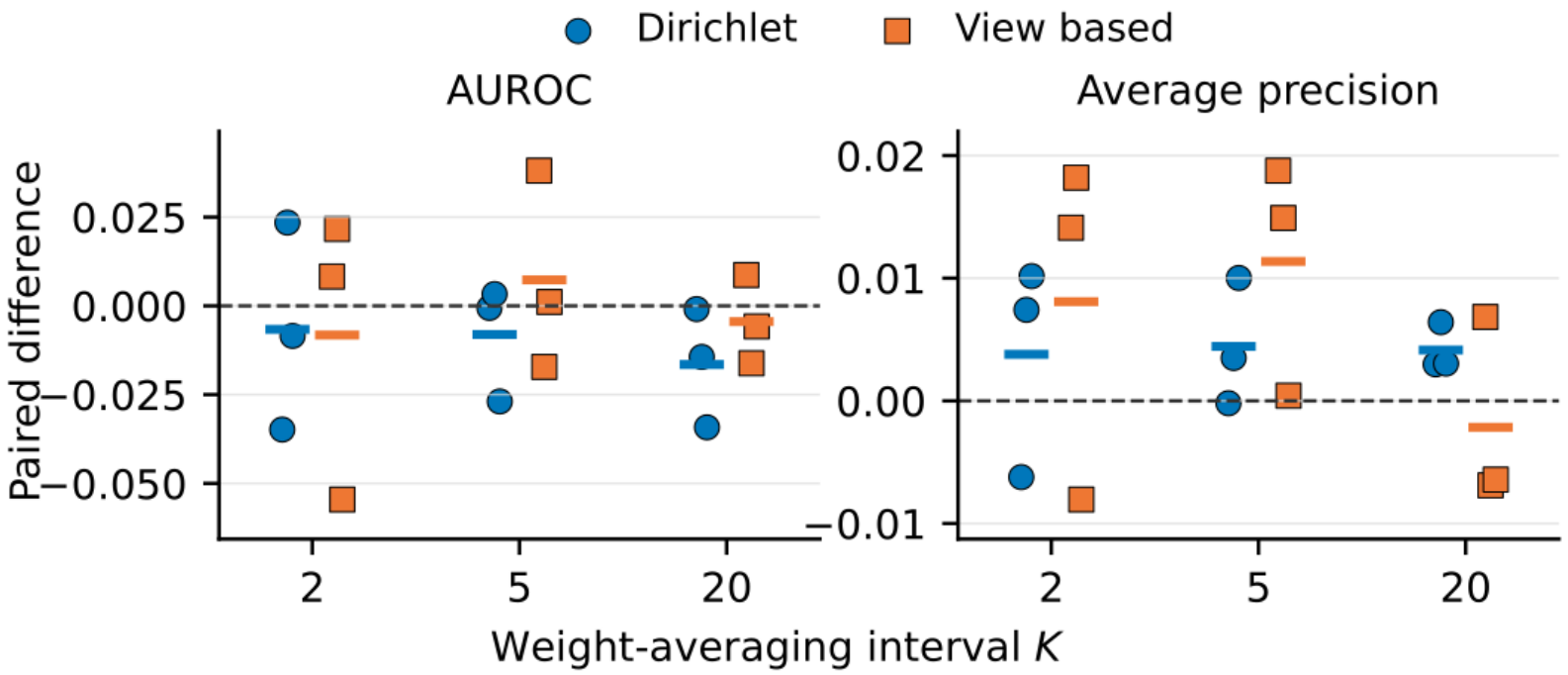
Comparison throughout: prototype alignment minus averaging-only control.

Prediction and geometry answer different questions

Prediction	Geometry	Interpretation
AUROC Average precision Validation checks	Effective rank Leading variance Within-class trace	Lower rank means concentration; it is not automatically a predictive gain.

This separation is the key to reading the results.

Predictive effects are setting-dependent



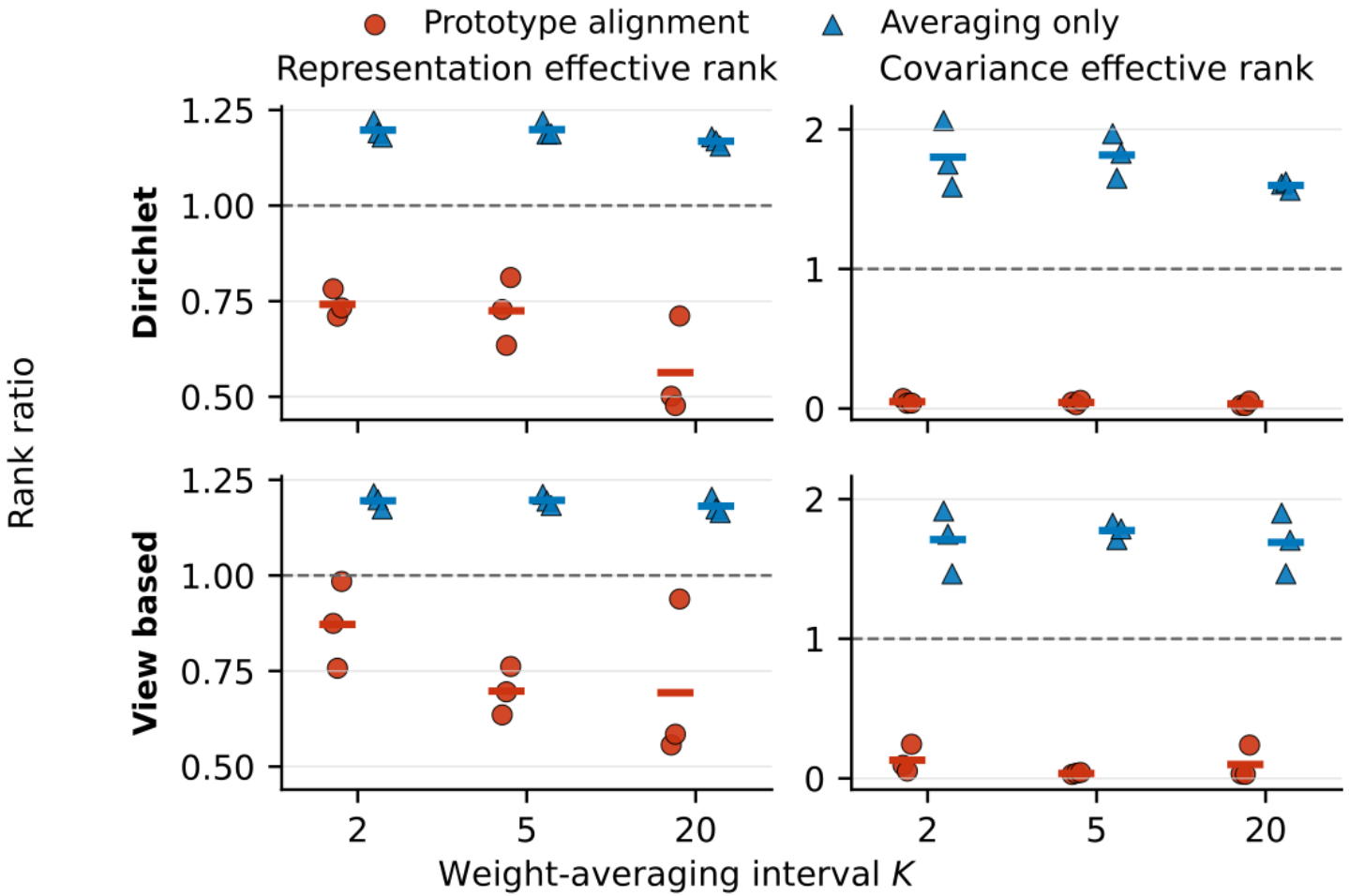
Across six matched settings

AUROC
-0.0165 to +0.0073

Average precision
-0.0022 to +0.0114

Prediction must be measured directly.

The spectral effect is consistent



Across every matched allocation and K

Representation rank decreases by 84-157

Covariance rank decreases by 114-129

Geometry moves more reliably than prediction.

The same concentration appears without federation

Pooled training

Aligned covariance rank: 1.42-1.58
Control: 119.93-126.62
Mean log-rank change: -4.4241

VinDr-CXR check

3,000 label-free images
All 21 pairs:
lower ranks, stronger leading direction

The alignment objective, not federation alone, induces the spectral signature.

A class mean is not a neutral summary of within-class variation.

1 **Prototype alignment concentrates centered representations.**

2 **Predictive effects vary by metric and setting.**

3 **Spectral and predictive endpoints should be reported separately.**

Scope: one binary CXR task, one ViT backbone, one alignment weight, three seeds.

Thank you.